

Actor–Critic methods

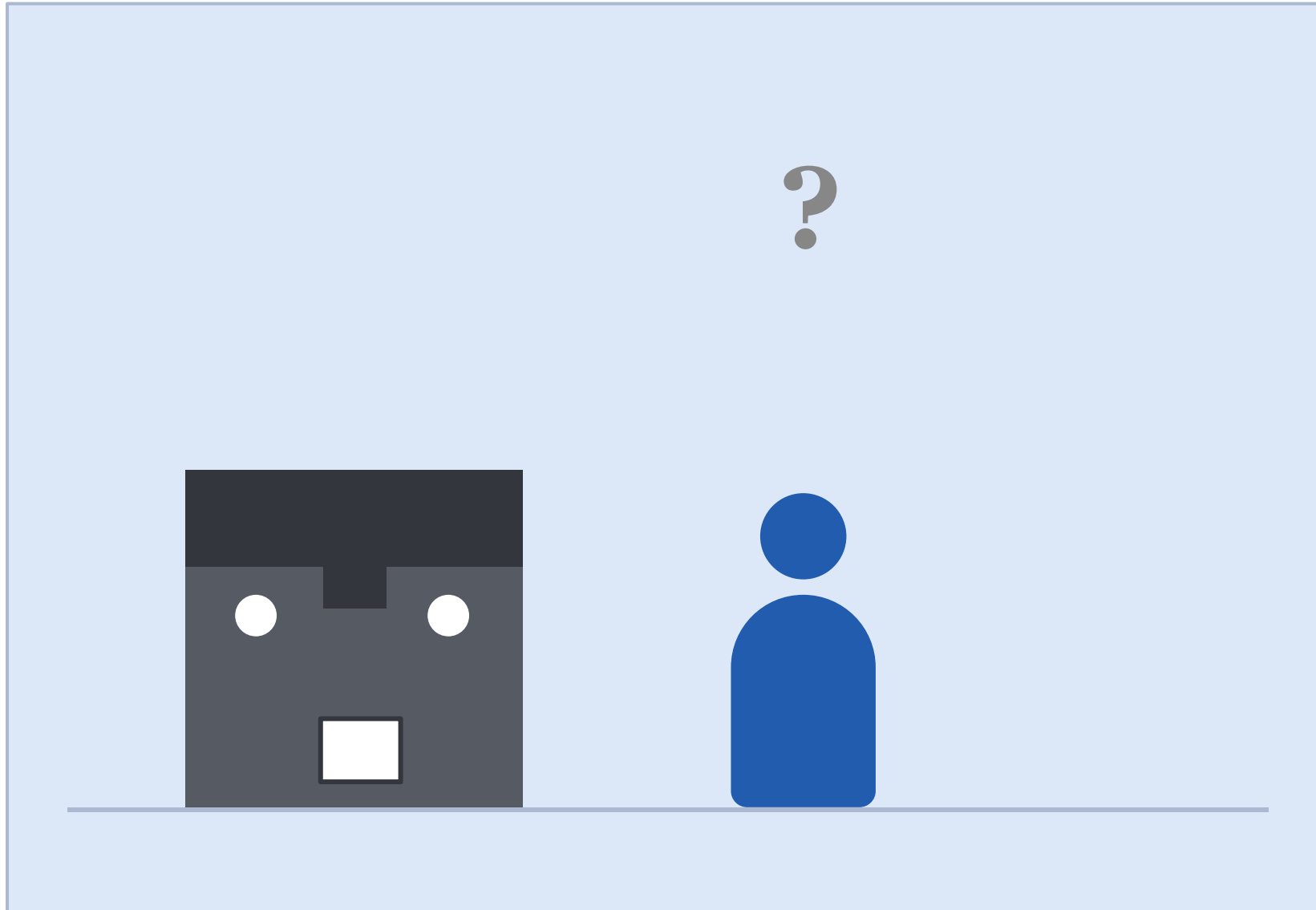
Learning to act and to evaluate

Mattia De Martino

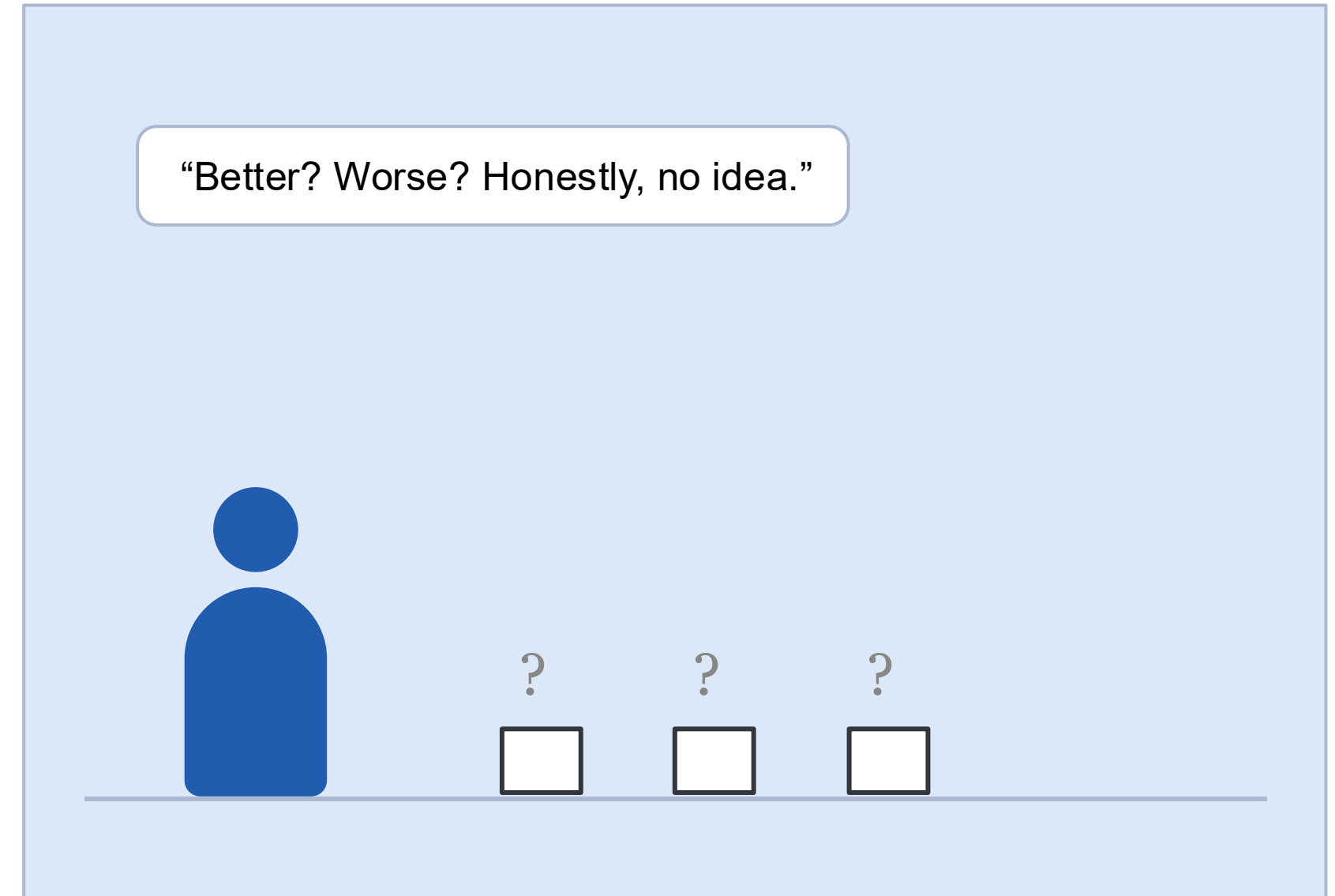
7.08.2026



A story to begin



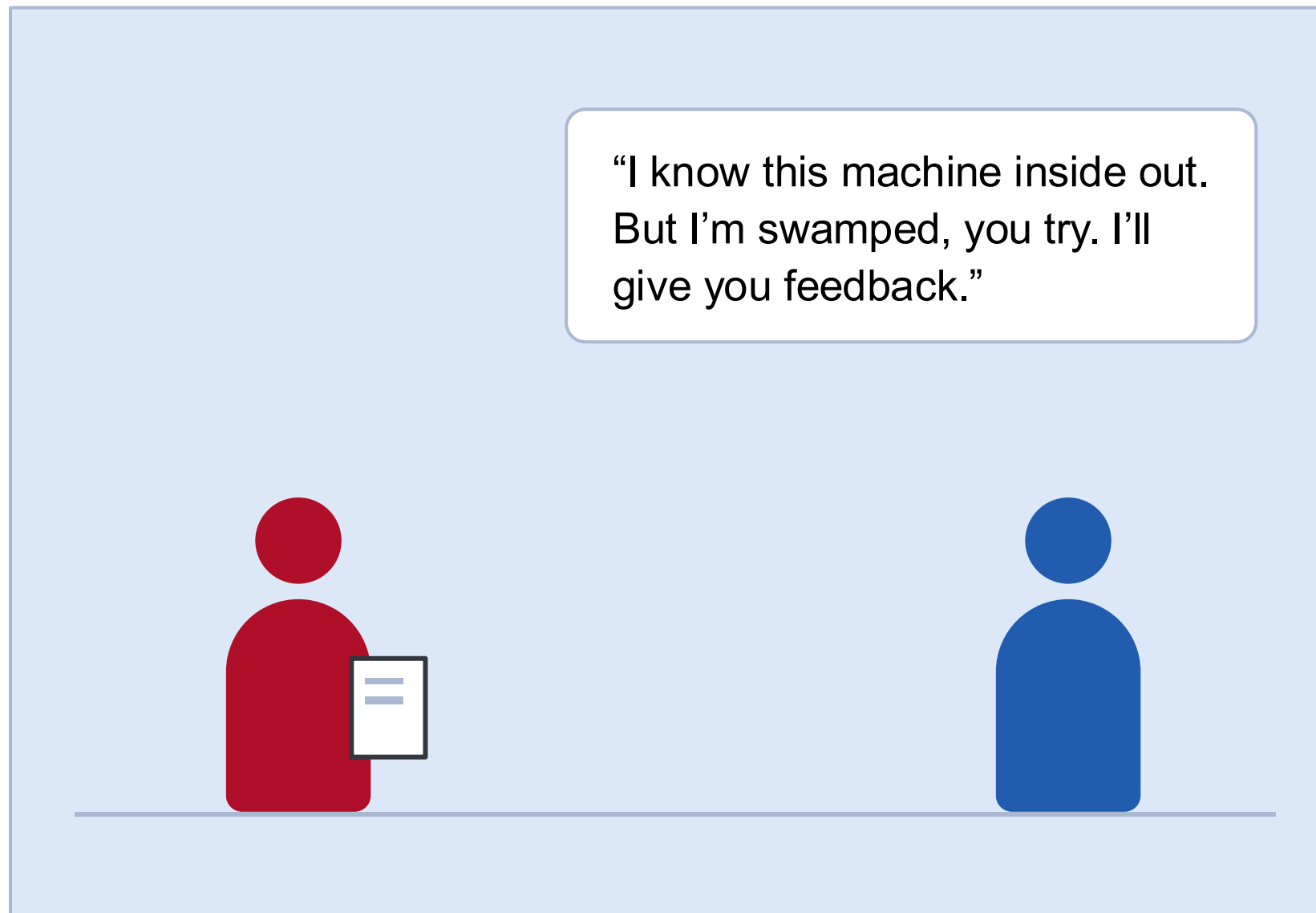
Antonio is handed a task he has never done: getting the office espresso machine to produce a decent cup.



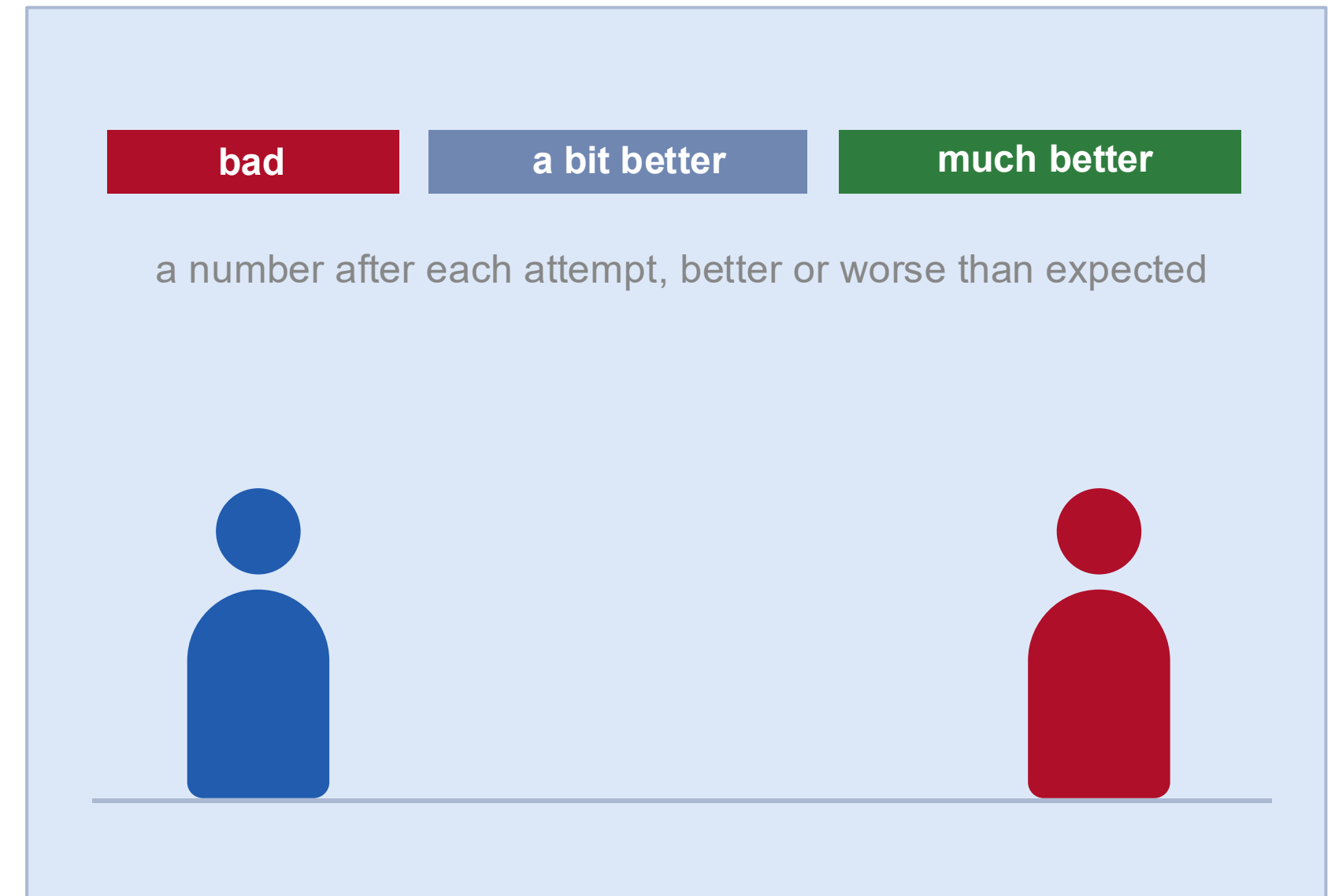
He experiments. He sees the outcome of every attempt but cannot tell which decision made it better or worse.

Some attempts do seem to work out better than others. Slowly, very slowly, he starts to learn.

A supervisor appears



Christian knows this machine well, he says. But he is far too busy, so the task stays with Antonio.



Now every action gets a graded score. Antonio leans away from what scored below expectation and toward what scored above and learns much faster.

The twist



The truth: Christian had never used that machine in his life. His first verdicts were guesses.

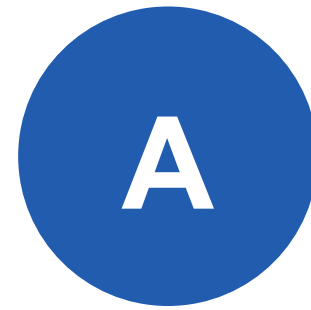


While Antonio experimented, Christian studied the outcomes, learning, attempt by attempt, to judge which actions were good.

One learned by trying. The other learned by observing and evaluating. Each got better thanks to the other.

About our two colleagues

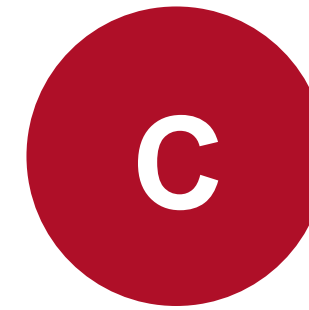
The story of Antonio and Christian matters a great deal in Reinforcement Learning. You may, in fact, have already heard their full names.



Antonio C. Tor



A. C. Tor



Christian Ritić

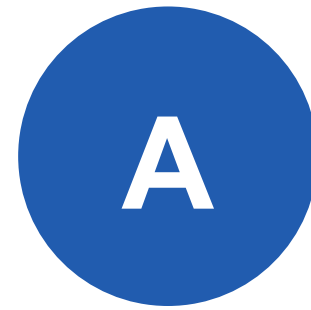


C. Ritić

The story you just heard is the story of this lecture, and of one of Reinforcement Learning's most important architectures.

About our two colleagues

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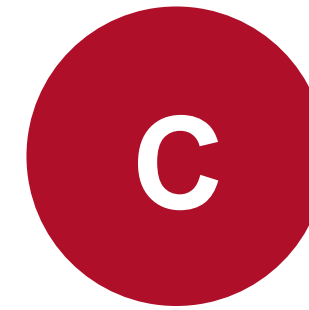


ACTOR



Predict the action from the state

What should I do here?



CRITIC



Gives feedback to the actor

Was that action good?

Today's lecture

- 1 **The story: what are learning?**
- 2 **Policy gradients: learn behaviour directly**
- 3 **Value and advantage: compare actions in the same state**
- 4 **The critic network: estimate $V(s)$ from experience**
- 5 **Training the system: update the critic, then the actor**
- 6 **Takeaways: one feedback loop to remember**

Policy search looks for the behaviour with the largest total return

The objective is long-term: choose a policy whose complete trajectories collect the most reward.

$$G(\tau) = r_0 + \gamma r_1 + \dots + \gamma^T r_T$$

- A policy chooses an action in every state.
- The return adds the rewards collected along the complete trajectory.
- Training changes the policy so high-return trajectories become more likely.

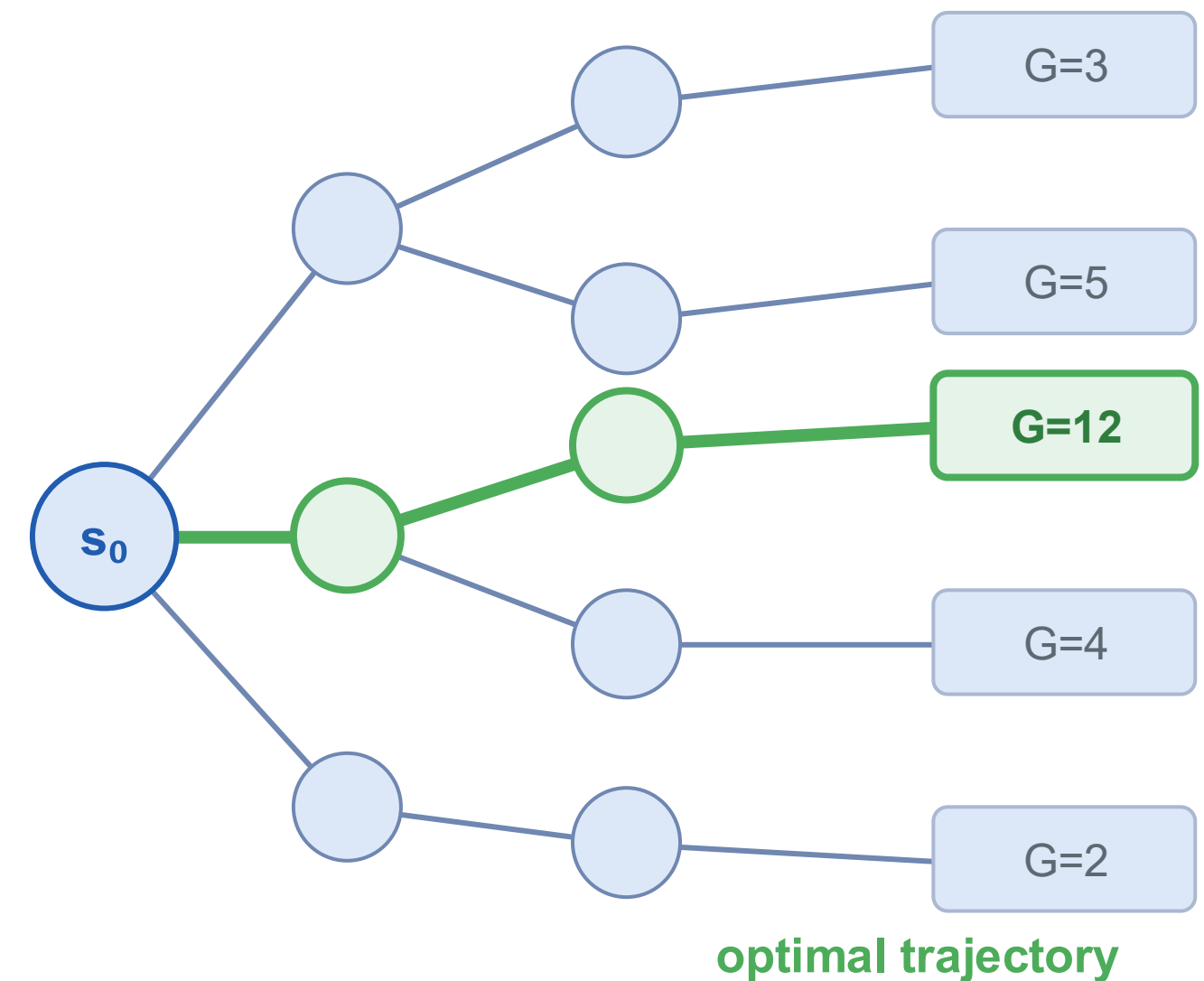


REINFORCE explores trajectories, then reinforces the best ones

Different action samples create different futures; one total return scores each complete trajectory.

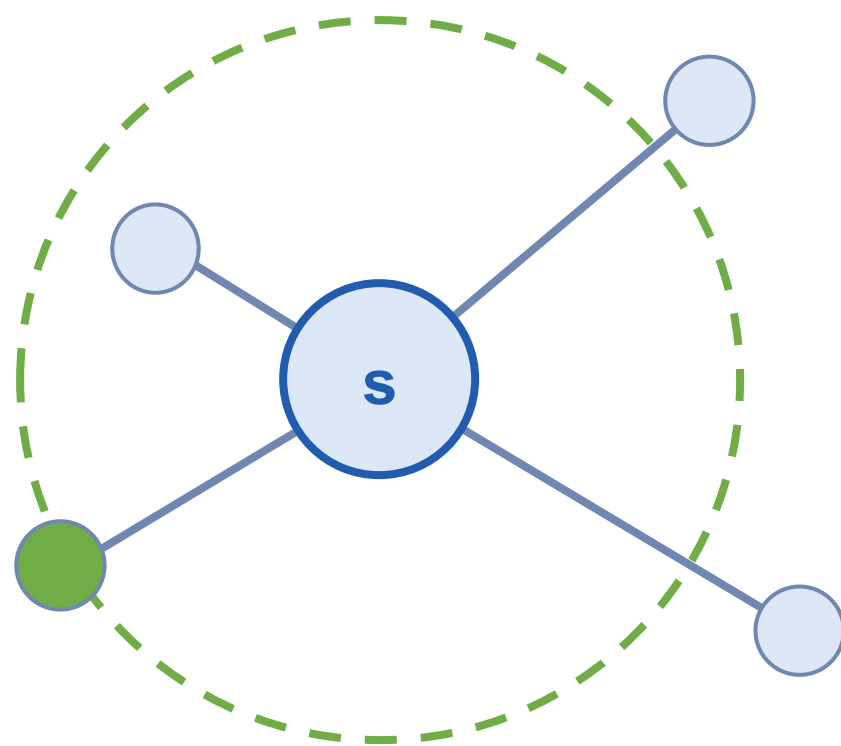
$$G(\tau) = r_0 + \gamma r_1 + \dots + \gamma^T r_T$$

- Every rollout follows a different trajectory through the environment.
- Each complete trajectory receives one total return.
- During training we try to make high-return trajectories more likely.



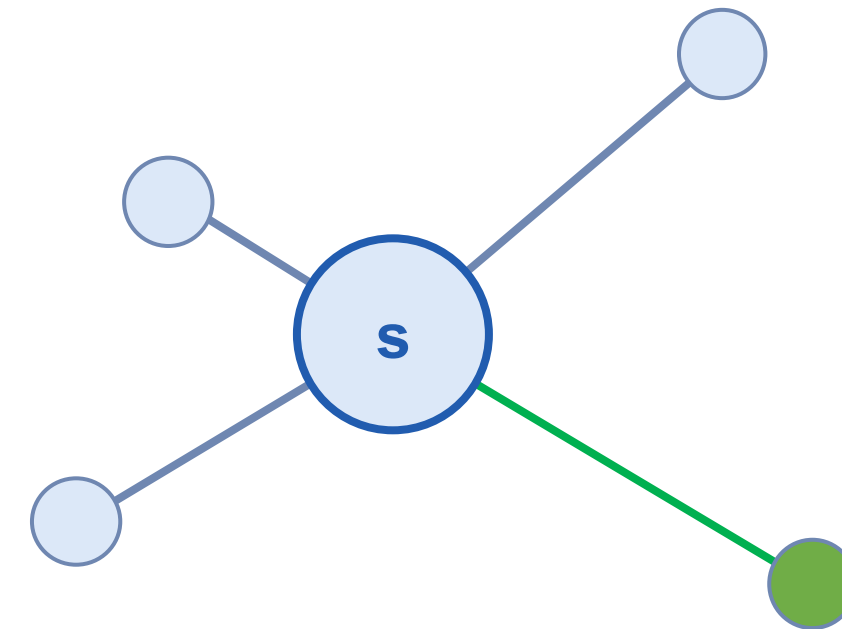
How much good is one state?

State value



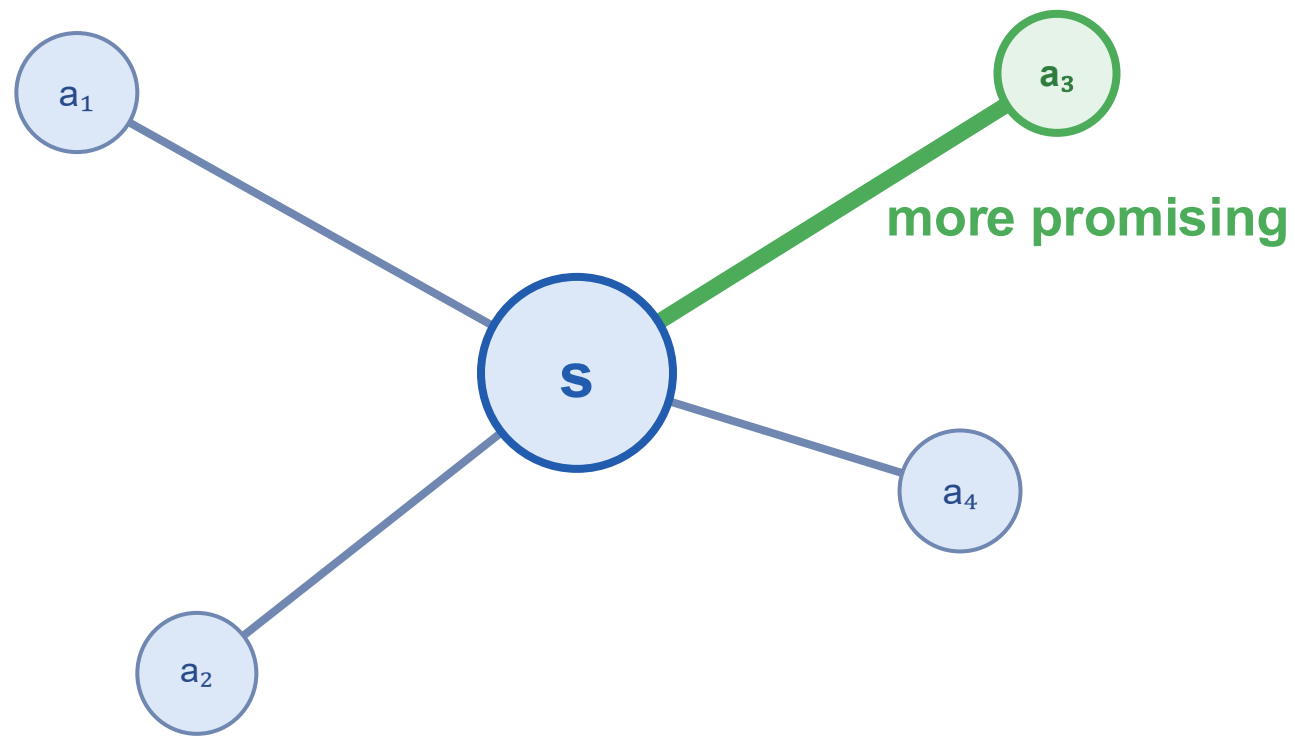
How promising is this situation

One action



How promising is doing this action in this state

How much good is an action?

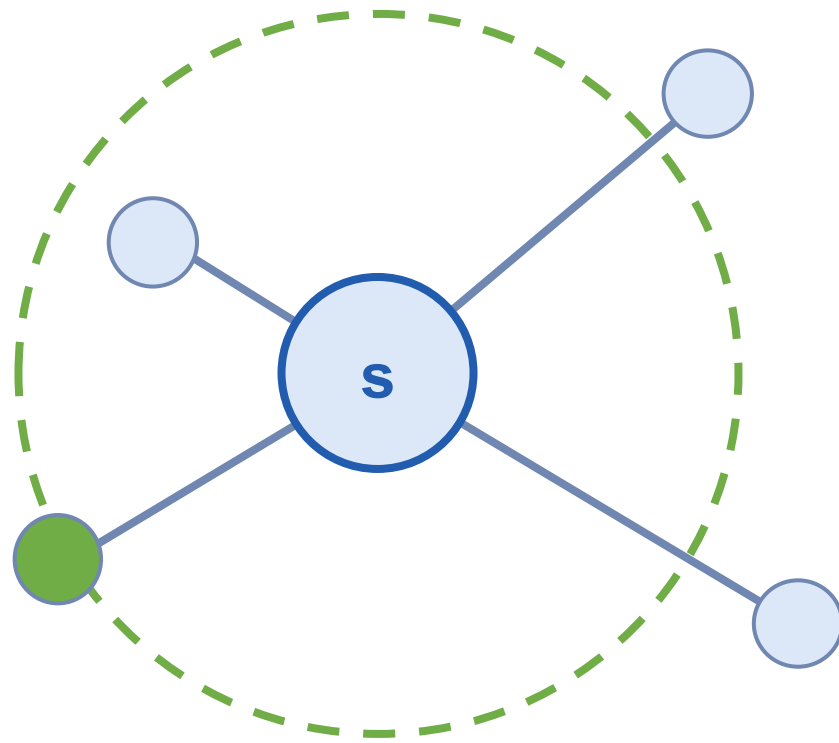


$$r(a, s) + \gamma V(s')$$

- $r(a, s)$: reward of doing our action in our state
- s' : the state I will reach doing the action
- $V(s')$: the value of the state s'

We define the value of an action as the sum of the reward and the discounted value of the next state

How much good is a state?



State value

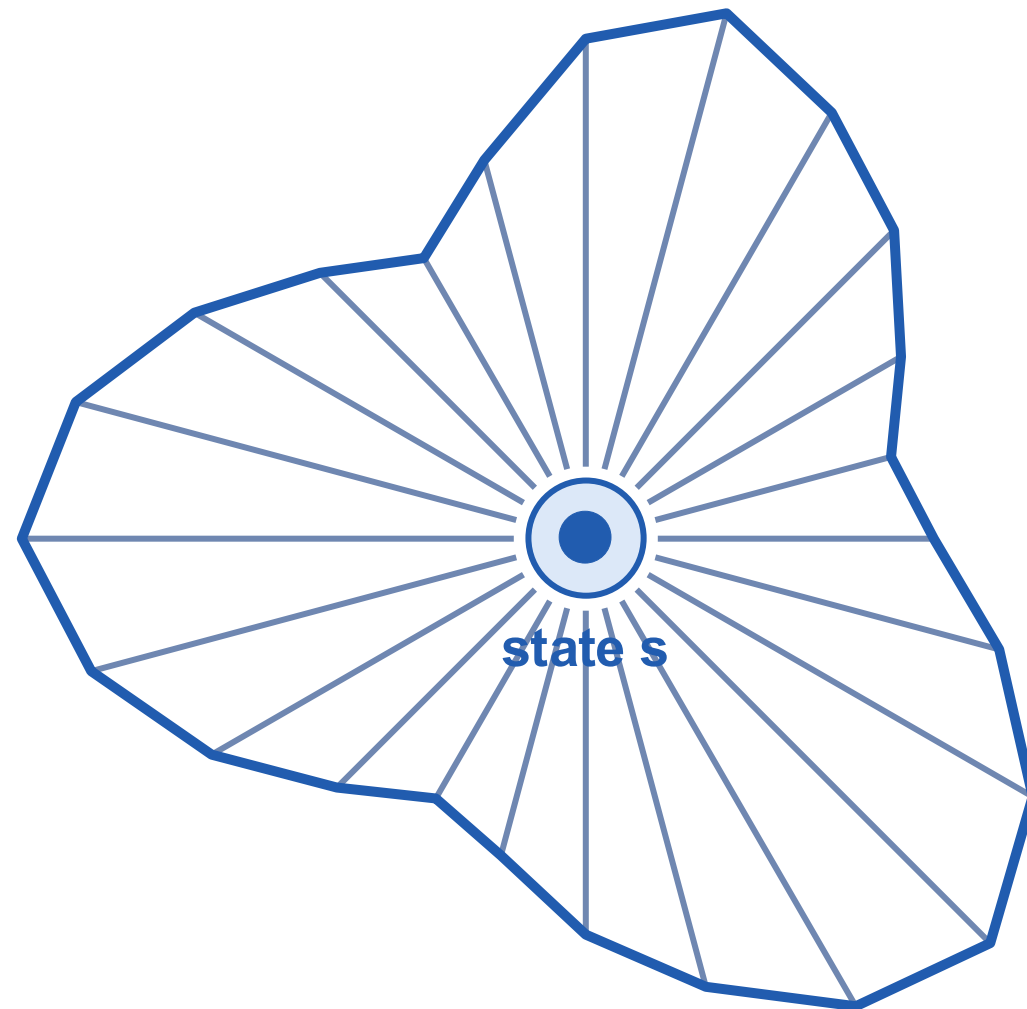
$$V(s) = r(a, s) + \gamma V(s')$$

- $r(a, s)$: reward of doing our action in our state
- s' : the state I will reach doing the action
- $V(s')$: the value of the state s'
- a : the action that we do following our policy

We define the state value as the expected return of being in a state and follow our policy

Each action leads to a different next state and a different value

Every direction is an action; distance represents its expected return.



— $r(a, s) + \gamma V(s')$

Expected return after choosing action a in state s .



What the rays reveal

Longer radius = more promising.
Shorter radius = less promising.

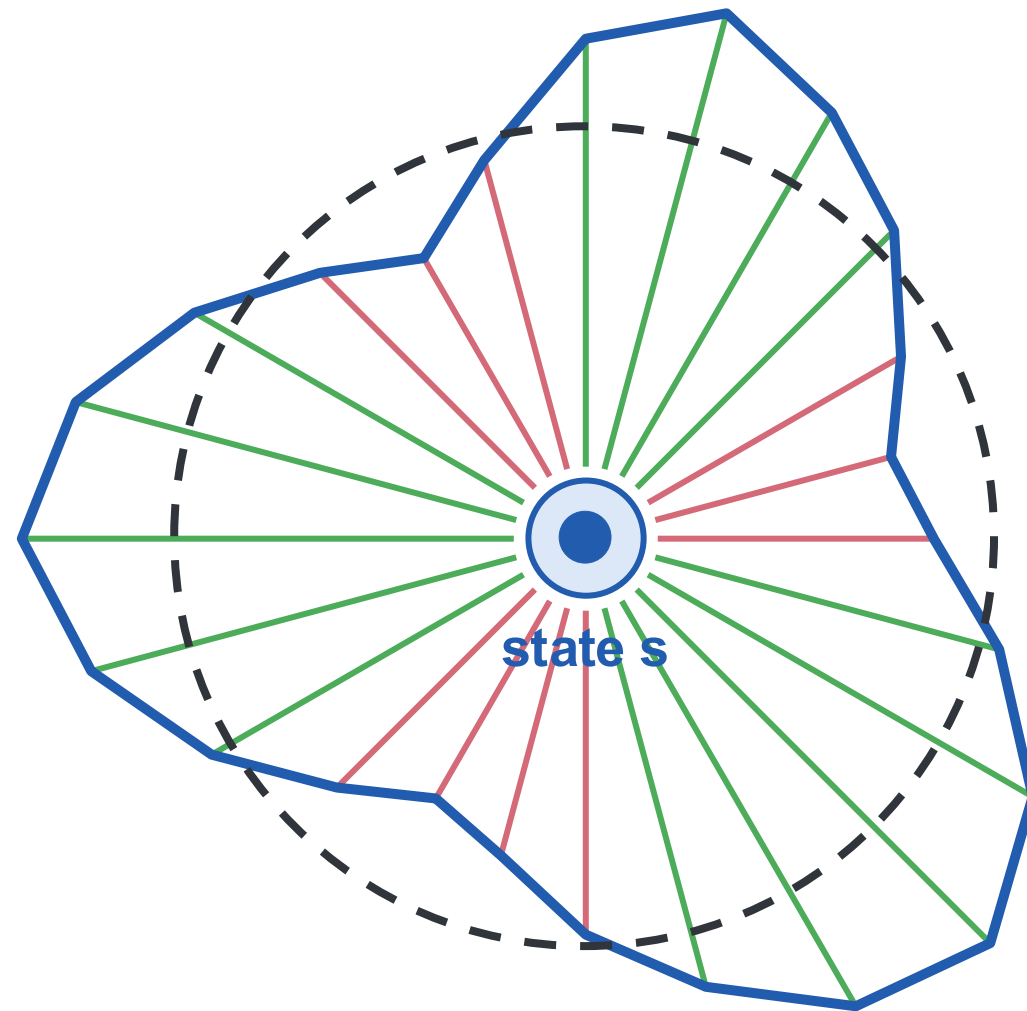
— **Best action = farthest point**

— **The rays rank the available actions**

Each ray shows what one action could achieve. V averages those values according to what our policy usually chooses.

Advantage is the gap between one action and the usual result

V is “what I usually get here”; $r + \gamma V(s')$ says whether this action is better or worse.



— $r(a, s) + \gamma V(s')$

The blue boundary: “what if I do this?”

○ $V(s)$

The circle: “what I usually get.”

— **above V → better than usual**

— **below V → worse than usual**

Advantage formalizes the comparison

Compare one action with the policy's usual result in the same state.

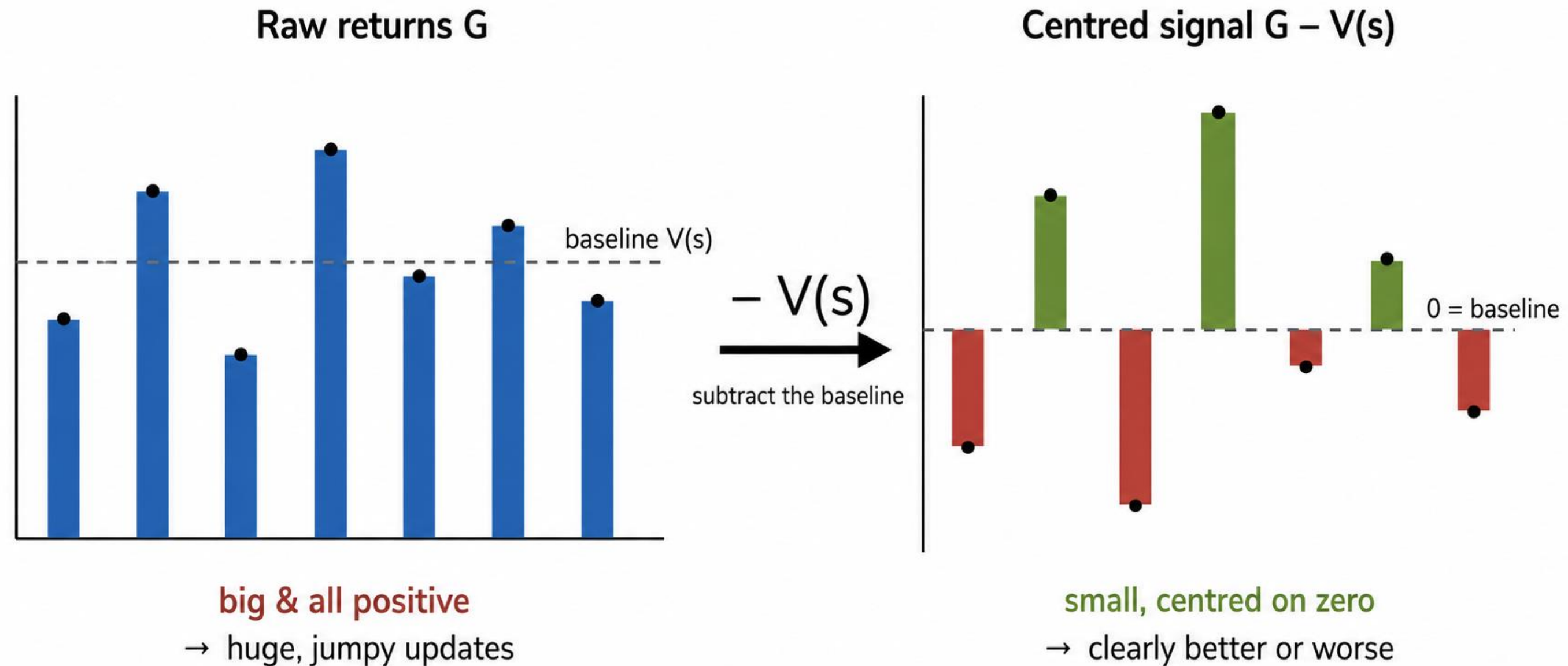
$$A(s, a) = r(a, s) + \gamma V(s') - V(s)$$

- s is the current state
- s' is the state that I will reach by doing the action
- r is the reward of doing that action



A baseline turns a score into a comparison

Subtract the expected return: the spread usually shrinks, while the learning direction stays.



The algorithm learns from better or worse than expected

$$\theta \leftarrow \theta + \alpha \underbrace{(G_t - b(s_t))}_{\hat{A}_t} \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t)$$

$$G_t > b(s_t) \implies \pi_{\theta}(a_t \mid s_t) \uparrow$$

$$G_t < b(s_t) \implies \pi_{\theta}(a_t \mid s_t) \downarrow$$

How can we estimate the state value?

The critic turns a state into one value estimate

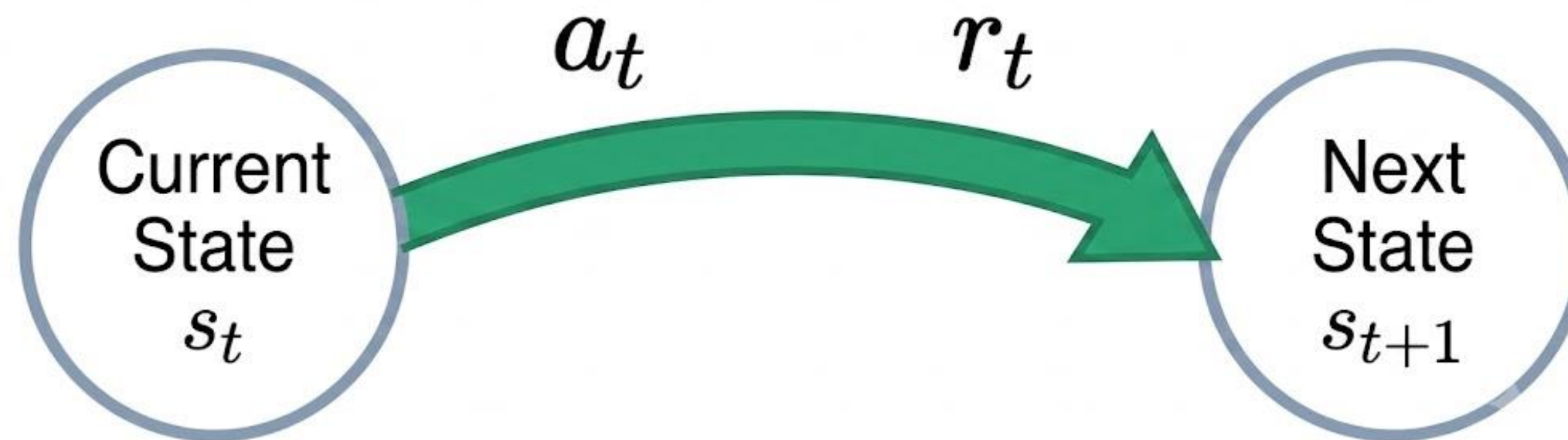
Input: the current situation. Output: the expected discounted return from here.



Build the critic's target

One observed reward plus one estimate of the remaining future.

$$y_t = r_t + \gamma V(s_{t+1})$$



Build the critic's target

One observed reward plus one estimate of the remaining future.

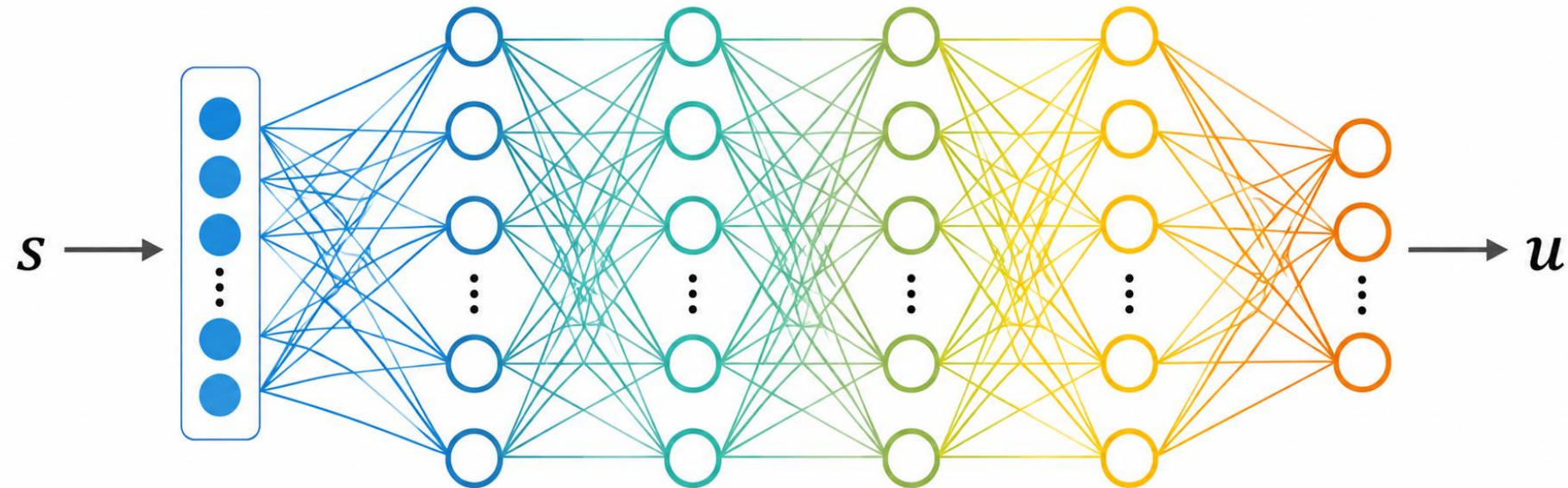
$$y_t = r_t + \gamma V(s_{t+1})$$

Recalling the **Bellman Equation**:

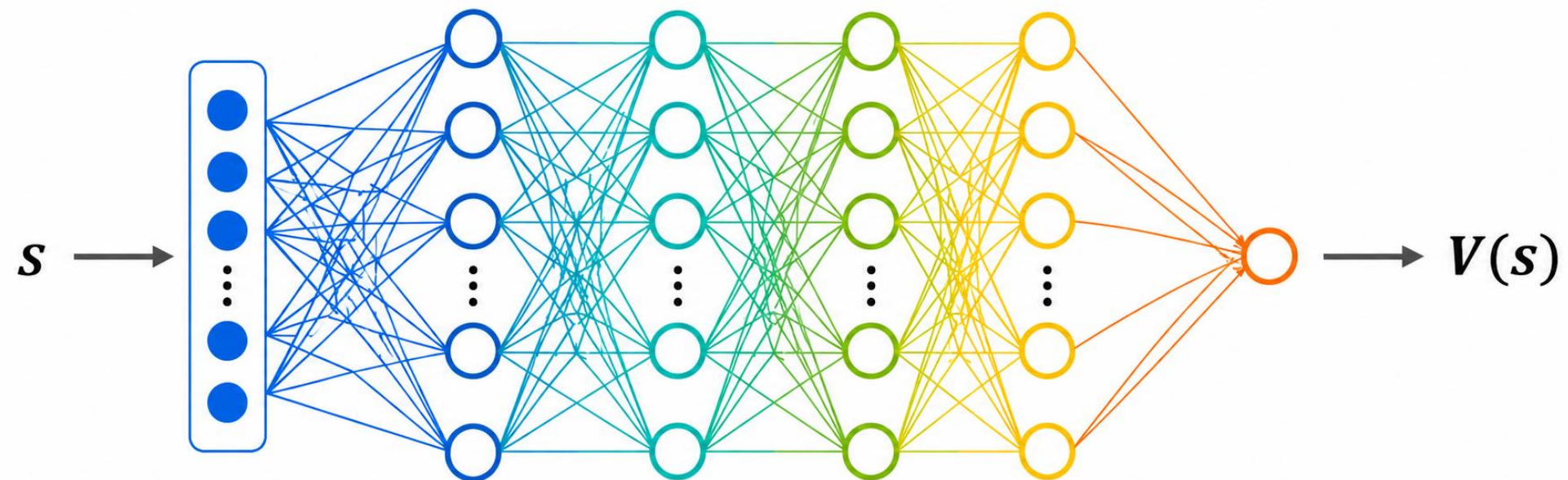
$$V^\pi(s) = r(a, s) + \gamma V^\pi(s')$$

Our two characters

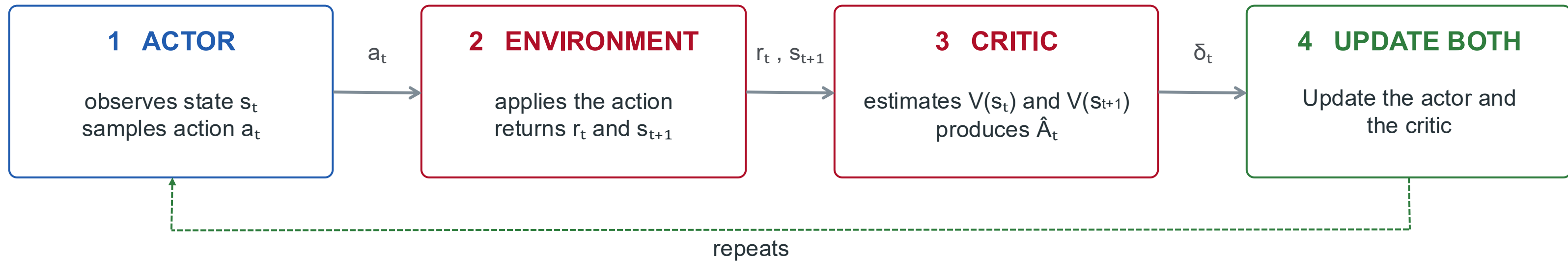
The actor



The critic



The complete Actor–Critic training loop



V-critic:

$$\mathcal{L}_c = \left(\overset{\text{target}}{\boxed{r_t + \gamma V_w(s_{t+1})}} - \overset{\text{output}}{\boxed{V_w(s_t)}} \right)^2$$

Critic update:

$$w \leftarrow w - \eta \nabla_w \mathcal{L}_c$$

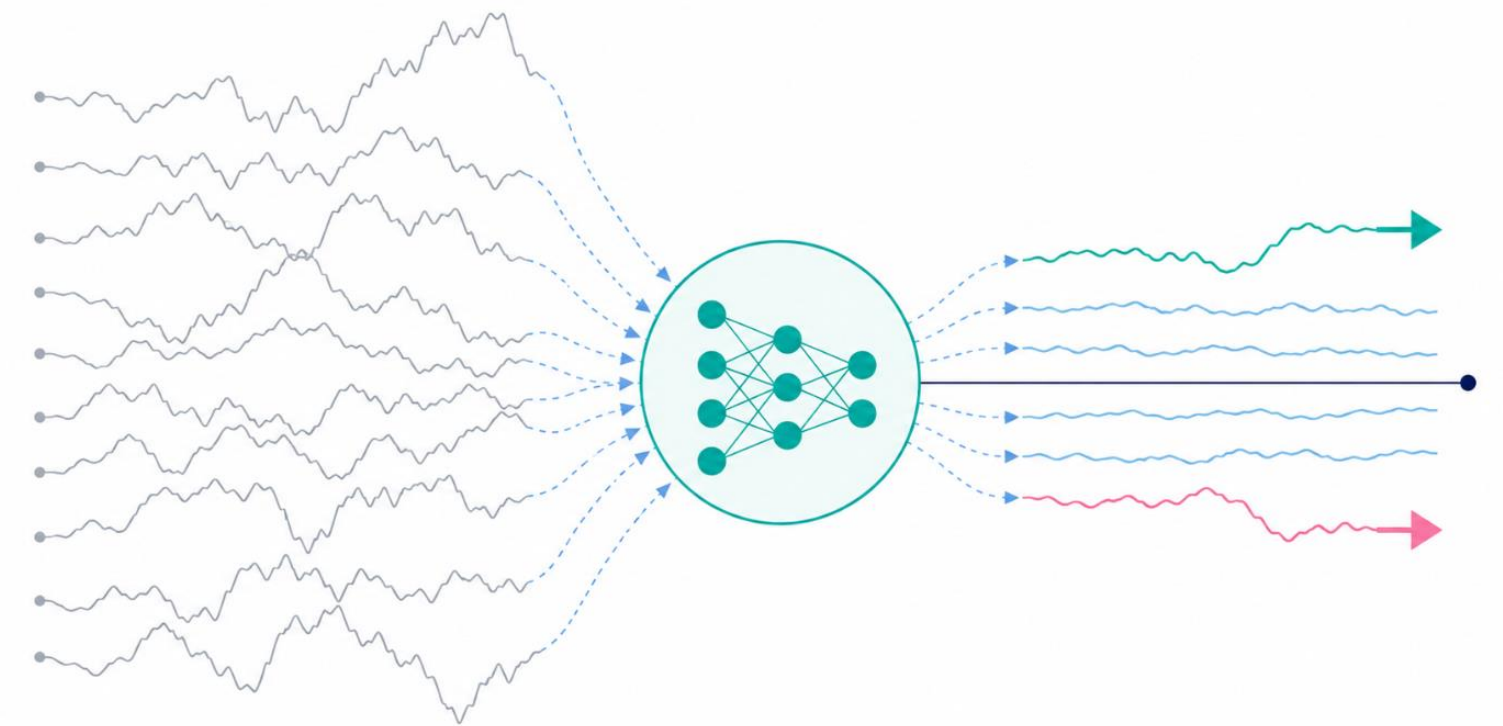
Actor update:

$$\theta \leftarrow \theta + \alpha \hat{A}_t \nabla_\theta \log \pi_\theta(a_t | s_t)$$

What we gain and what we trade

A learned critic improves the signal, but it is not a free oracle.

- **Gain:** lower-variance updates than raw REINFORCE returns.
- **Gain:** learning after every transition, not only at the end of an episode.
- **Trade:** a wrong critic introduces bias and can destabilize the actor.



Practical rule: keep actor and critic learning at compatible speeds, so the policy does not chase a rapidly moving estimate.

What to take home

- 1 The Actor chooses the actions, while the Critic evaluates the decisions taken.
- 2 The Actor does not learn only from the reward, but from whether the outcome was better or worse than expected.
- 3 The Critic improves its predictions, while the Actor uses those predictions to improve the policy
- 4 The Actor and the Critic learn together and depend on each other

Questions?